

CHEMICAL SIGNALS AND KIN RECOGNITION

IN SPINY MICE (Acomys cahirinus)

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It follows from kin selection theory that the ability to influence the transmission of one's alleles into future generations would be enhanced in organisms who can readily distinguish between close relatives and unrelated conspecifics (Hamilton, 1964). Recognition of kin allows for preferential allocation of resources to such individuals (i.e., nepotism), which in turn increases the likelihood of survival and successful reproduction by the recipients--who share a large proportion of alleles with their related benefactor (e.g., Alexander, 1979). Likewise, optimal choice of mates, for example, reaching an appropriate balance between extreme outbreeding and incestuous mating, could be facilitated by discerning close kin (Bateson, 1983; Shields, 1983).

Kin recognition is believed to be a significant factor in the evolution of social behavior and has therefore become the subject of considerable current research (for recent reviews see Holmes and Sherman, 1983; Porter, in press). It appears that the sensory modalities involved in the recognition of kin are those that are otherwise the most salient for communication within a particular species. Accordingly, chemical cues have been implicated in the mediation of kin recognition in diverse taxa, including frog tadpoles (Blaustein and O'Hara, 1982; Waldman, 1984), coho salmon (Quinn and Busack, 1985), a variety of rodent species (e.g., Porter et al., 1978b; D'Udine and Partridge, 1981; Holmes, 1984), and even humans (Macfarlane, 1975; Russell, 1976; Porter and Moore, 1981; see also Colgan, 1983). The remainder of this chapter will focus upon kin recognition in spiny mice, Acomys cahirinus (a murid rodent indigenous to the eastern Mediterranean region); with particular emphasis on the role of chemical signals in parent-infant and sibling recognition.

PARENT-INFANT RECOGNITION

The most frequently studied version of kin recognition is that involving parents and offspring. Genetic benefits accruing to parents who

respond preferentially to their own offspring are fairly obvious. Postnatal (or post-hatching) investment in progeny has direct positive consequences on the propagation of one's genes. In contrast, individuals who indiscriminately distribute their parental resources incur the genetic costs of wasted investment in unrelated young and, concomitantly, depriving their own offspring of their limited resources (Trivers, 1972). Thus, in contexts where biological offspring are likely to co-occur with unrelated young there would be considerable pressure favoring the evolution of mechanisms enabling parents to recognize their own offspring.

Within the first two days postpartum, laboratory-reared female spiny mice will indiscriminately nurse and retrieve their own offspring and unfamiliar unrelated pups of the same age (Porter and Doane, 1978). When two mothers, along with their eight-day-old pups are placed together, however, nursing interactions most often involve mothers and their own pups (Porter et al., 1980a). Related investigations indicate that the indiscriminate maternal responsiveness of one- to two-day postpartum females may be due (at least in part) to the similarity of chemical cues associated with newborn pups whose mothers are maintained on the same laboratory diet. Recently parturient females more rapidly retrieve unfamiliar pups born of females fed the same diet as the experimental mothers in comparison to simultaneously present pups born to females fed a distinctively different diet (Doane and Porter, 1978). Furthermore, the survival rate of pups fostered onto females fed the same diet as the mothers of the foster pups is greater than that of pups fostered onto different-diet females. Chemical cues that mediate maternal recognition of offspring thus appear to be dietary-dependent, so that the odors of neonates born to (and nursed by) mothers maintained on identical diets would be somewhat similar, whereas pups of mothers consuming different diets would have discernibly distinctive odors.

Spiny mice, like other muroid rodents, have eclectic feeding habits--taking advantage of whatever food sources are available. The diets of wild-living animals would presumably fluctuate and the types of food ingested, as well as the relative proportions of similar food items, differ somewhat across individuals. Therefore, reliance on chemical cues (labels) associated with the mother's unique diet would be a reliable strategy for early maternal recognition of offspring. Although the source of the dietary-dependent pup-recognition labels is not known at present, such cues could be deposited directly on the infants by the mother (e.g., through her saliva while grooming the pups, or through her urine), or result from the ingestion of mother's milk.

Unlike females, male spiny mice begin to respond discriminatively to their own familiar offspring vs. alien pups--all of whom were born of females maintained on the same diet--within 48 hours of birth (Makin and Porter, 1984). Such paternal recognition of offspring appears to involve chemosensory signals since males tested with two simultaneously present pups lick and sniff an alien pup more than their own. The increased responsiveness to the alien pups has been interpreted as heightened chemosensory investigation of an unfamiliar stimulus rather than a preference for such neonates over the fathers' own offspring.

Mammalian young are entirely dependent upon willing adult conspecifics (typically their mother) for the provision of nutrients and, possibly, other